

WEATHERWISE — Multi-Tenant Weather AI SaaS Platform

Cloud API Gateway Architecture, Subscription Billing Engine & B2B SaaS Ecosystem

1. EXECUTIVE SUMMARY & SAAS PLATFORM VISION

WeatherWise is a multi-tenant B2B/B2C Software-as-a-Service (SaaS) weather intelligence platform engineered for cloud-native deployment. Built on a decoupled microservices architecture, WeatherWise transforms raw meteorological feeds into high-value contextual insights. While conventional consumer weather applications render static numbers, WeatherWise operates an intelligent SaaS backend with an integrated **API Gateway, Rate-Limiting Engine, Tenant Isolation, and Automated Stripe Subscription Billing**.

The platform provides a dual value proposition: a subscription-based web client for consumers receiving AI-driven lifestyle/attire advice, and a metering-based Developer API SaaS allowing B2B enterprises (logistics, agriculture, event platforms) to embed weather advisories into their own business workflows with strict SLA-backed uptime.

2. ENGINEERING ROSTER & PROJECT OWNERSHIP

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COURSE MODULE

Web Application Engineering & SaaS Architecture

DEPLOYMENT STACK

AWS Serverless (Lambda, API Gateway, DynamoDB)

REPOSITORY SOURCE

MahadRazaKhan/ai-based-weather-app

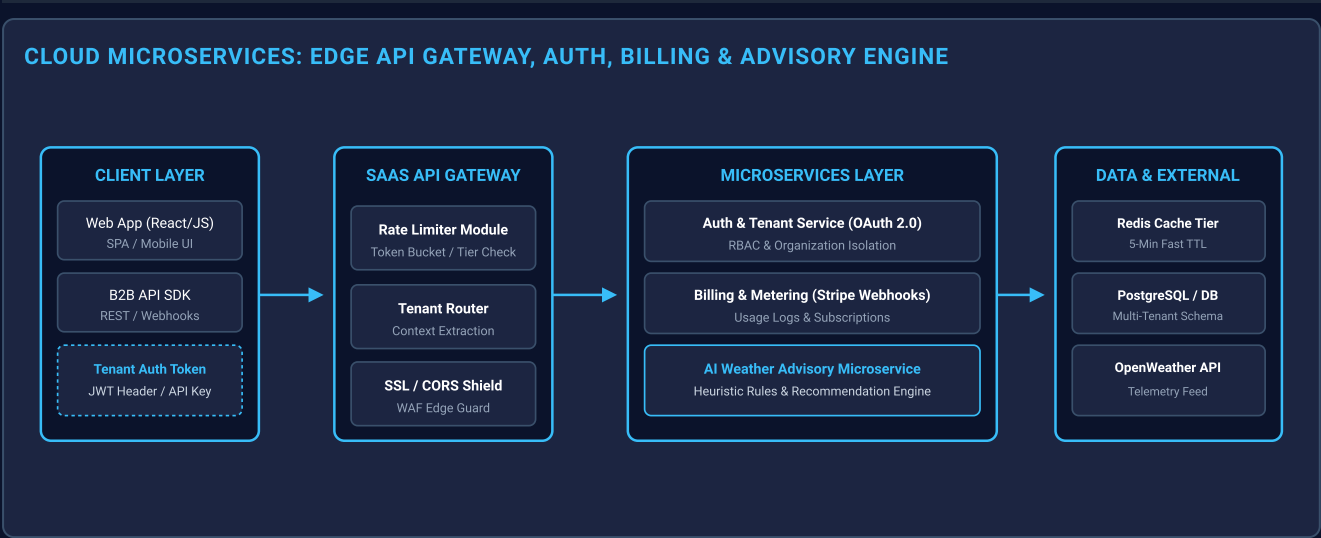
MONETIZATION MODEL

Freemium / B2B SaaS Subscription (Stripe Billing)

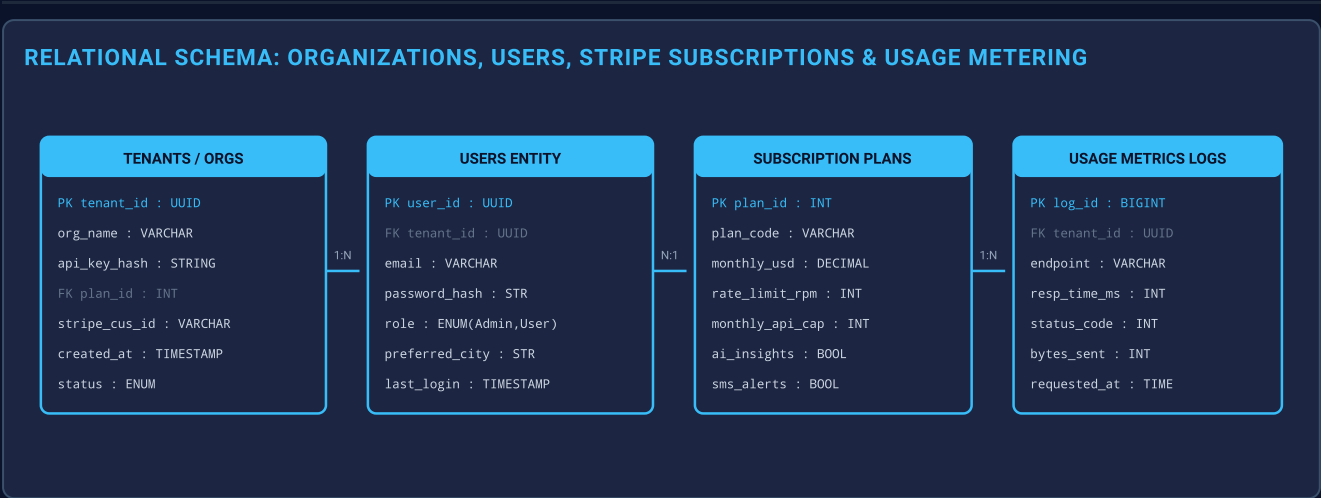
CORE SAAS DIFFERENTIATION

WeatherWise leverages tenant-aware caching (Redis) and serverless API metering. It offers zero-friction onboarding, automated plan upgrades via Stripe Webhooks, and granular Role-Based Access Control (RBAC) across Free, Pro, and Enterprise tiers.

3. MULTI-TENANT SAAS SYSTEM ARCHITECTURE



4. MULTI-TENANT SAAS ENTITY-RELATIONSHIP (ER) SCHEMA



5. SAAS CORE CAPABILITIES & MULTI-TENANT FEATURES

Tenant-Aware API Middleware API Gateway validates API keys and JWTs, applying dynamic token-bucket rate limiting based on the organization's subscription plan.	AI Context Advisory Engine Transforms temperature and humidity into rule-based lifestyle guidance (attire recommendations, outdoor activity suitability metrics).
Multi-Region Redis Caching Reduces OpenWeather API calls by 85% via 5-minute spatial Redis caching, minimizing cloud egress charges and lowering latency.	Stripe Subscriptions & Webhooks Automated subscription lifecycle management handling upgrades, downgrades, payment retries, and instant plan provisioning.

6. PRODUCTION SAAS TECHNOLOGY STACK

LAYER	TECHNOLOGIES	SAAS ARCHITECTURAL PURPOSE
Frontend UI	React.js, Tailwind CSS, HTML5	Single Page Application (SPA) with responsive tenant management dashboards.
API Gateway	AWS API Gateway, Express / Node	JWT authentication, SSL termination, dynamic rate-limiting, and CORS management.
Microservices	Node.js (ES6+), Python FastAPI	Stateless backend microservices handling auth, weather heuristic rules, and billing.
Databases & Cache	PostgreSQL, Redis	Relational multi-tenant schemas (Row-Level Security) with high-speed key-value cache.
Billing & DevOps	Stripe API, Docker / AWS Lambda	Automated billing, usage metering, and serverless auto-scaling deployment.

7. SAAS MONETIZATION & SUBSCRIPTION TIERS

PLAN TIER	TARGET AUDIENCE	PRICING	QUOTA & SAAS SLA FEATURES
Developer Free	Casual Users & Devs	\$0 / Month	1,000 API calls/mo, standard 5-day forecast, basic attire suggestions, community support.
Pro SaaS	Power Users & SMBs	\$29 / Month	100,000 API calls/mo, 60 RPM rate limit, SMS/WhatsApp severe weather alerts, ad-free UI.
Enterprise B2B	Logistics & Agri Fleet	\$299 / Month	Unlimited API access, 99.9% uptime SLA, dedicated Redis cache tenant, 24/7 priority support.

8. SAAS EXECUTION & SCALING ROADMAP

- Phase 1 (Production MVP):** Multi-tenant React frontend, OpenWeather integration, basic rule-based AI recommendations, and responsive UI.
- Phase 2 (SaaS Billing & Auth):** OAuth 2.0 / JWT integration, Stripe subscription webhook pipelines, and PostgreSQL tenant schema migration.
- Phase 3 (Enterprise Scale):** Global CDN distribution, Redis cluster deployment, automated B2B usage metering dashboard, and LLM-powered advisory extensions.